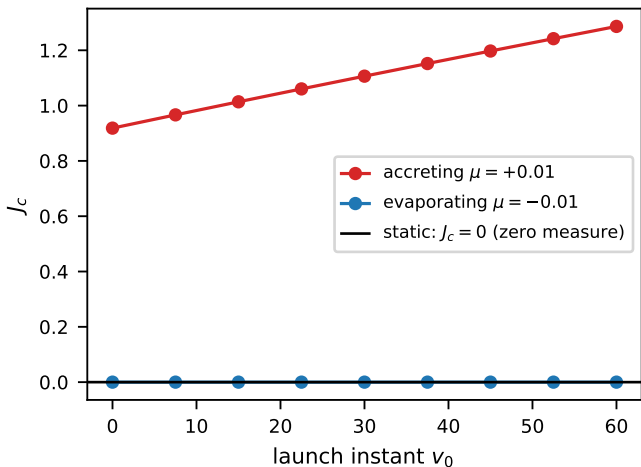


penetration threshold $J_c(v_0)$: below J_c the τ -brachistochrone crosses $r = 2m(v)$ ($E = 1.2$, $r_0 = 10$)



horizon chase near the threshold
($\mu = +0.01$, $v_0 = 0$, $J_c = 0.9185$)

